Audio Amplifier Testing Procedure

Testing Plan A

Table 1: Testing plan for maximum power output

Item	Details
Objective	Measure the maximum power output, P_{MAX}, for a total harmonic distortion (THD) below 1% using the first five harmonics. Engineering requirements: • THD < 1% for an output power of 1W, up to the 5^{th} harmonic.
Assumptions	• 1kHz input sinewave • 8Ω load • 12V supply
Hazards and PPE	Hazards: • Electric shock Personal Protective Equipment: • Safety glasses • Insulated gloves
Tools required	• Oscilloscope • Function generator • DC power supply
Test Setup, Required Environment	The test should be completed in room temperature (20°C to 25°C) conditions. Setup: • Set the volume (potentiometer) to max. • Set the DC power supply at 12V constant voltage with a 2A current limit. • Set the function generator in High Z. • Set the function generator to a 440mV_{pp} sinewave at 1kHz and zero offset. • Set the oscilloscope acquisition mode to averaging. • Connect the oscilloscope to the input and output of the amplifier using BNC probes. • Connect the DC power supply to V_{ss} on the amplifier using a banana-hook. • Connect the function generator to the input of the amplifier using a BNC-hook.

Procedure (Execution and Data Collection)	1. Display the Fast Fourier Transform (FFT) on the output waveform, setting center to 3kHz and span to 5kHz. 2. Turn on the DC power supply and function generator output. 3. Adjust the oscilloscope to fit around 20 periods of the waveform. 4. With the cursor, measure the peak dB of each harmonic, record V_{HARM} into page A of the Audio Amplifier Performance Calculator. 5. The calculated THD should be close but less than 1% (0.9% < THD <1%). Adjust V_{pp} on the function generator if needed, continue till the correct THD is obtained. Record this THD value into Table 2. 6. With the cursor, measure the peak voltage of the input and output waveforms, record VI_{MAX} and VO_{MAX} into Table 2 and page A of the Audio Amplifier Performance Calculator. 7. Repeat till eight trials are completed.
Data Analysis & Conclusion	The Total Harmonic Distortion (THD) measures distortions by comparing the true harmonic with the distorting harmonics. For calculating Total Harmonic Distortion: $$THD = \frac{\sqrt{V_{2k-RMS} + V_{3k-RMS} + V_{4k-RMS} + V_{5k-RMS}}}{V_{1k-RMS}} * 100$$ To calculate the max power output, use the equation: $$P_{MAX} = \frac{VO_{MAX}^2}{R_L}$$ The load of the amplifier, R_L, is assumed to be 8 Ω. The Max Power Output must be greater or equal to 1W for the test to be successful.

Table 2: Maximum power output data collection sheet

Trial #	THD (%)	VI_{MAX} (V_{pp})	VO_{MAX} (V_{pp})	Max Power Output (W)	Pass/Fail
1					
2					
3					
4					
5					
6					
7					
8					

Tester Signature: ______________________________________

Date of Test: ____

Testing Plan B

Table 3: Testing plan for frequency response

Item	Details
Objective	Measure the frequency response of the audio amplifier from 20Hz to 20kHz. Create a bode plot in dB vs. log frequency and angle vs. log frequency. Engineering requirements: • The gain over 100Hz and 20kHz should be consistent (± 3dB) with an 8Ω load. • The max voltage gain should be at least 26dB.
Assumptions	• $0.5*VI_{MAX}$ V_{pp} input • 8Ω load • 12V supply
Hazards and PPE	Hazards: • Electric shock Personal Protective Equipment: • Safety glasses • Insulated gloves
Tools required	• Oscilloscope • Function generator • DC power supply
Test Setup, Required Environment	The test should be completed in room temperature (20°C to 25°C) conditions. Setup: • Set the volume (potentiometer) to max. • Set the DC power supply at 12V constant voltage with a 2A current limit. • Set the function generator in High Z. • Set the function generator to a peak-to-peak voltage of $0.5*VI_{MAX}$ sinewave with zero offset. (Obtain VI_{MAX} from Testing Plan A) • Set the oscilloscope acquisition mode to high resolution. • Connect the oscilloscope to the input and output of the amplifier using BNC probes. • Connect the DC power supply to V_{ss} on the amplifier using a banana-hook. • Connect the function generator to the input of the amplifier using a BNC-hook.
Procedure (Execution and Data Collection)	1. Set the function generator to 20 Hz and measure V_{IN}, V_{OUT}, and phase by using quick measure. Record these values into page B of the Audio Amplifier Performance Calculator. (Phase is output waveform → input waveform) 2. Increase the frequency of the function generator, measuring at least every octave till 20kHz. In the Audio Amplifier Performance Calculator and Table 2, preset frequencies are determined.

Data Analysis & Conclusion	For calculating gain (dB): $$Gain\ (dB) = 20 * \log\left(\frac{V_{OUT}}{V_{IN}}\right)$$ The highest gain calculated in Table 4 will be Max Gain. There are two requirements for this test to be successful. The max gain must be larger than 26dB. The gain from 100Hz to 20kHz must not be below 3dB from the Max Gain.

Table 4: Frequency response data collection sheet

Frequency (Hz)	Gain (dB)	Phase (°)	Pass/Fail
20			
40			
80			
160			
320			
640			
1280			
2560			
5120			
10240			
20000			

Tester Signature: ____

Date of Test: ____

Testing Plan C

Table 5: Testing plan for I/O impedance

Item	Details
Objective	Measure the input and output impedance of the audio amplifier, with a 1kHz input. Engineering requirements: • The AC input impedance should be larger than 1kΩ • The AC output impedance should be less than 1Ω
Assumptions	• 1kHz input • 0.5*VI_{MAX} V_{pp} input • 12V supply
Hazards and PPE	Hazards: • Electric shock Personal Protective Equipment: • Safety glasses • Insulated gloves
Tools required	• Function generator • DC power supply • Digital multimeter
Test Setup, Required Environment	The test should be completed in room temperature (20°C to 25°C) conditions. Setup: • Set the volume (potentiometer) to max. • Set the DC power supply at 12V constant voltage with a 2A current limit. • Set the function generator in High Z. • Set the function generator to a peak-to-peak voltage of 0.5*VI_{MAX}, 1kHz sinewave with zero offset. • Connect the DC power supply to V_{ss} on the amplifier using a banana-hook. • Connect the function generator to the input of the amplifier using a BNC-hook.

Procedure (Execution and Data Collection)	Input Impedance: 1. Use a load of 4Ω for the audio amplifier. 2. Take an input AC voltage measurement using the digital multimeter, probing the input of the amplifier parallel to ground. Record the value in page C of the Audio Amplifier Performance Calculator. 3. Take an input AC current measurement using the digital multimeter, probing the function generator probe in series with the input probe of the audio amplifier. Record the value in page C the Audio Amplifier Performance Calculator. Output Impedance 1. Use a load of 4Ω for the audio amplifier 2. Take an output AC voltage measurement using the digital multimeter, probing the output of the amplifier parallel with ground. Record the value in page C of the Audio Amplifier Performance Calculator. 3. Take an output AC current measurement using the digital multimeter, probing the output of the amplifier in series with the load resistor. Record the value in page C of the Audio Amplifier Performance Calculator. 4. Repeat steps 2-3 using a 16Ω resistor. Repeat whole procedure till eight trials are completed.
Data Analysis & Conclusion	To calculate R_{IN} (Ω), use the equation: $$R_{IN} = \frac{V_{IN-4}}{I_{IN-4}}$$ To calculate R_{OUT} (Ω), use the equation: $$R_{OUT} = \left\lvert\frac{(V_{OUT-16} - V_{OUT-4})}{(I_{OUT-16} - I_{OUT-4})}\right\rvert$$ There are two requirements for the test to be successful: • R_{IN} (Ω) > 1kΩ • R_{OUT} (Ω) < 1Ω

Table 6: I/O impedance data collection sheet

Trial #	R_{IN} (Ω)	R_{OUT} (Ω)	Pass/Fail
1			
2			
3			
4			
5			
6			
7			
8			

Tester Signature: ______________________________________

Date of Test: ______________________________________

Testing Plan D

Table 7: Testing plan for power efficiency

Item	Details
Objective	Measure the audio amplifier power efficiency for different input levels. Engineering requirements: • Efficiency should be greater than 40% at P_{MAX} with a 1kHz input, 8Ω load, and 12V supply
Assumptions	• 1kHz input sinewave • 8Ω load • 12V supply
Hazards and PPE	Hazards: • Electric shock Personal Protective Equipment: • Safety glasses • Insulated gloves
Tools required	• Function generator • DC power supply • Digital multimeter
Test Setup, Required Environment	The test should be completed in room temperature (20°C to 25°C) conditions. Setup: • Set the volume (potentiometer) to max. • Set the DC power supply at 12V constant voltage with a 2A current limit. • Set the function generator in High Z. • Connect the DC power supply to V_{ss} on the amplifier using a banana-hook. • Connect the function generator to the input of the amplifier using a BNC-hook.
Procedure (Execution and Data Collection)	1. Set the peak-to-peak voltage of the function generator to VI_{MAX} (V_{pp}). 2. Take the power supply AC voltage measurement using the digital multimeter, probing the power supply AC voltage of the amplifier parallel to ground. Record the value in page D of the Audio Amplifier Performance Calculator. 3. Take the power supply AC current measurement using the digital multimeter, probing the DC power supply in series with the power rail input, V_{SS}, of the audio amplifier. Record the value in page D of the Audio Amplifier Performance Calculator. 4. Take an output AC voltage measurement using the digital multimeter, probing the output of the amplifier parallel with ground. Record the value in page D of the Audio Amplifier Performance Calculator. 5. Take an output AC current measurement using the digital multimeter, probing the output of the amplifier in series with the load resistor. Record the value in page D of the Audio Amplifier Performance Calculator. 6. Repeat for a function generator peak-to-peak voltage of $0.5*VI_{MAX}$.

Data Analysis & Conclusion	To calculate the power for both the amplifier and load, use the equation: $$P = VI$$ The efficiency is found by: $$Eff.(\%) = P_{OUT}/P_{SS}$$ P_{OUT} is the power delivered to the 8Ω load and P_{SS} is the power sourced from the power supply. For the test to be successful, efficiency must be greater than 40% using a V_{IN} of VI_{MAX}.

Table 8: Power efficiency data collection sheet

V_{IN} (V)	V_{SS} (V)	I_{SS} (A)	V_{OUT} (V)	I_{OUT} (A)	Eff. (%)	Pass/Fail
VI_{MAX}						
$0.5*VI_{MAX}$						

Tester Signature: ____

Date of Test: ____